

## Introduction to Human-Computer Interaction

### What is Human-Computer Interaction (HCI)?

**Human-Computer Interaction (HCI)** is a cross-disciplinary area that deals with the theory, design, implementation, and evaluation of ways human use and interact with computing devices. HCI uses productivity, safety, and entertainment to support and fulfill human-computer activities. It initially focused on computer interface design and has now expanded to cover almost all forms of information technology design.

HCI has become more important in recent years as computers have become part of a human's daily routine. Aside from making the necessary computational functionalities available, its focus has been on how to design and implement computer interfaces for high usability and user experience.

- **High usability** is the interactive design and operations through specific computer system criteria which includes correctness, efficiency, safety, utility, learning, and retention.
- **User experience** focuses on creating systems that enhance the overall involvement of a person, which must include a satisfying, enjoyable, entertaining, helpful, motivating, aesthetically pleasing, and fun experience to the user.

Humans interact with computing devices through a user interface. This includes software, such as those being displayed on a computer monitor or any mobile device, and hardware, such as keyboard, mouse, and other external devices.

- **Interaction** refers to an abstract model by which humans interact with the computing device to complete a given task.
- **Interface**, on the other hand, refers to the boundary wherein two (2) independent systems meet and act on or communicate with each other.

### Importance of HCI

Human-computer interaction is important. Usable and efficient interaction with different computing devices translates to higher productivity in any industry. A poor interface would make it difficult for users to benefit from even the simplest systems or programs.

The simple aesthetic appeal of an interface, while satisfying the need for usability, is now a critical requirement for commercial success. Creating products and services that work seamlessly is important because it gives any company the edge and therefore sells better. One good example is the family of distinctly designed Apple products. Apple products are very attractive and have created a crowd of regular customers who patronize their products; even its functionalities are nearly equal to their competitors.

HCI had a significant impact on the history of computing and changed human's daily lives. Without even knowing where it exactly started, HCI still continues to redefine how to view, absorb, exchange, create, and manipulate information technology towards human advantage. Below are some noticeable impacts:

- The spreadsheet interface made business computing a success.
- The Internet phenomenon could not have happened without the Web-browser interface.
- Smartphones and other mobile devices are widely used today.
- Body-based and action-oriented interfaces are now being used to play and enjoy computer games.

### Principles of HCI

#### Know the User

Commonly known as "user-centered design," this principle serves as the primary principle of HCI. This implies that the interaction and interface should cater to the needs and capabilities of the target users of the system in design. Basic information, such as gender, age, education level, and computing experience, are collected

and analyzed to determine possible preferences, tendencies, physical and mental capabilities, and skill levels of users. This information can then serve as a basis to choose the right mode of interaction and design the right interface solution for the target users appropriately.

Examples:

- Profiling the target user of a program within a company
- Knowing the age bracket of the software users in an educational institution
- Conducting a survey about the computing experience of the employees in a company who will utilize the system

Contrary to this, the concept of *universal usability* caters to a wide range of users rather than specific ones. This concept promotes humane interfaces. As simple as it sounds, the universal usability concept is generally difficult to achieve and implement on a single system interface because of the extensive requirements in the advanced multicultural societies around the world.

### Understand the Task

Understanding the task is another plausible principle of HCI. The term “task” refers to the job to be accomplished by the user with the interactive system. This principle is closely related to interaction modeling and user analysis. Understanding the task encapsulates sequence identification and structure of subtasks at an abstracted level that are appropriate for users within larger application context.

Examples:

- Formulating an interaction model for the task of logging into a website
- Creating a conceptual model for system access in a corporation
- Analyzing the user’s capability to navigate at a certain module of a program

### Reduce Memory Load

Humans are more efficient in performing tasks that require *light memory load*, either for long term or short term memory. Keeping the user’s short term memory load light is important in relation to the interface’s role as a quick and easy guide to complete a task. The capacity of human’s short term memory is about five (5) to nine (9) specific information with respect to the completion of a task. Light memory load also leads to less erroneous behavior.

Examples:

- Keeping the number of menu items or tabs to a lesser quantity than what is required for a human’s short term memory to maintain good user awareness of the ongoing task
- Windows and Mac provide users with keyboard shortcuts for copy and pasting and for users to navigate and operate more quickly and effortlessly.

### Strive for Consistency

A way to relieve long term memory load of a user is to strive for consistency. This applies to both interaction model and interface implementation. A user has a higher tendency to be confused and exhibit inaccurate and inappropriate responses if the same subtask is involved at a different time and for a different interaction step or interface methods. Aside from being able to remember what to do, consistency and familiarity lead to higher satisfaction and preference while using the system.

Examples:

- A consistent look of the interface within a gaming application
- A consistent interface between Microsoft PowerPoint and Microsoft Word

### Remind Users

A significant task involves the use of memory. Another strategy is to employ interfaces that provide constant reminders about important information or actions to be performed and thus refresh the user's memory. *Feedback* is associated with this principle. Feedback is an event that occurs when the output of a system is used as an input back into the system as part of a chain of events.

Examples:

- Confirming the safe receipt of a credit card number in an online payment
- Reminding the user to perform a system update on a computer

### Prevent Errors/Reversal of Action

In performing any given task, it is important to complete the task without any error. Therefore, the interaction and the interface should be designed to avoid confusion and overwork of users.

Examples:

- Confirming the user's action and displaying large interactive processes through pop-up menus
- Presenting only the relevant information at a given time and making selections rather than enforcing recall or full manual input specification by the user
- Making the user comfortable by allowing an easy reversal of action like the Undo button

### Natural Interaction and Interface

The last HCI principle is to favor natural interaction and interfaces. This principle refers to the reflection of various actions and processes in human's daily life. At some point, it can be complicated to translate real-life actions and other modes of interaction to interact with a computer directly, but it is still achievable.

Examples:

- The natural interactive interface of Wii
- The natural interactive interface of the online game Pokémon GO

---

### References:

- Beal, V. (2019). *Interface*. Retrieved from <https://www.webopedia.com/TERM/I/interface.html> on May 8, 2019
- Feedback. (n.d.). In *Techopedia*. Retrieved from <https://www.techopedia.com/definition/7159/feedback> on May 8, 2019
- Human-Computer Interaction (HCI). (n.d.). In *Interaction Design Foundation*. Retrieved from [https://www.interaction-design.org/literature/topics/human-computer-interaction?order\\_by=date](https://www.interaction-design.org/literature/topics/human-computer-interaction?order_by=date) on May 9, 2019
- Human-Computer Interaction (HCI). (n.d.). In *Techopedia*. Retrieved from <https://www.techopedia.com/definition/3639/human-computer-interaction-hci> on May 8, 2019
- Kim, G. (2015). *Human-computer interaction fundamentals and practice*. USA: CRC Press.
- Wong, E. (2019, May). *Shneiderman's eight golden rules will help you design better interface*. Retrieved from <https://www.interaction-design.org/literature/article/shneiderman-s-eight-golden-rules-will-help-you-design-better-interfaces> on June 6, 2019